

A.

Round Robin is a CPU scheduling method where each process receives a fixed amount of CPU time called a time quantum. After a process uses its assigned time, it moves to the back of the queue if it is not yet finished.

Imagine waiting in line to buy food in a cafeteria. Instead of one person taking unlimited time at the counter, each person gets a fixed amount of time to order. If they need more time, they return to the end of the line and wait for another turn. This ensures fairness and prevents one person from holding up everyone else.

Why is this scheduling algorithm important in computing systems?

Round Robin is important because it allows multiple tasks to share CPU resources fairly. It prevents long processes from blocking shorter ones and improves system responsiveness, especially in systems where users interact directly with computers.

B.

Round Robin Scheduling is a preemptive CPU scheduling algorithm where processes are executed for a limited amount of time called a time quantum.

Turnaround Time

The total time from arrival until completion.

Time Quantum

Fixed CPU time assigned to each process.

Process

A task or program currently being executed.

Burst Time

The amount of CPU time a process needs to complete.

Waiting Time

The total time a process spends waiting in the queue.

C.

Consider the following table of arrival time and burst time for three processes P1, P2 and P3 and given Time Quantum = 2 ms

Step-by-Step Execution:

Process	Burst Time	Arrival Time
P1	4 ms	0 ms
P2	5 ms	0 ms
P3	3 ms	0 ms

Time 0-2 (P1): P1 runs for 2 ms (total time left: 2 ms).

Time 2-4 (P2): P2 runs for 2 ms (total time left: 3 ms).

Time 4-6 (P3): P3 runs for 2 ms (total time left: 1 ms).

Time 6-8 (P1): P1 finishes its last 2 ms.

Time 8-10 (P2): P2 runs for another 2 ms (total time left: 1 ms).

Time 10-11 (P3): P3 finishes its last 1 ms.

Time 11-12 (P2): P2 finishes its last 1 ms.

Now, lets calculate average waiting time and turn around time:

Processes	AT	BT	CT	TAT	WT
P1	0	4	8	$8 - 0 = 8$	$8 - 4 = 4$
P2	0	5	12	$12 - 0 = 12$	$12 - 5 = 7$
P3	0	3	11	$11 - 0 = 11$	$11 - 3 = 8$

Turnaround Time = Completion Time - Arrival Time

Waiting Time = Turnaround Time - Burst Time

D.

Advantages

Fair allocation of CPU time

No process suffers starvation

Good response time

Suitable for time-sharing systems

Disadvantages

Too many context switches may reduce performance

Choosing the wrong time quantum can create problems

Large quantum behaves like FCFS

Small quantum causes excessive overhead

E.

Operating Systems

Examples include:

CPU process scheduling

Task scheduling in multi-user systems

Everyday Computing

Examples:

Multiple browser tabs running simultaneously

Smartphones handling background applications

Multiplayer online game servers

Printing queues

F.

Let us compare Round Robin with First Come First Serve (FCFS).

Feature	Round Robin	FCFS
Scheduling Type	Preemptive	Non-preemptive
Fairness	High	Low
Response Time	Faster	Slower
Starvation	No	Possible
Complexity	Moderate	Simple

Round Robin provides better responsiveness than FCFS because processes do not wait until one process completely finishes.

G.

To conclude, Round Robin Scheduling is a preemptive algorithm that assigns a fixed time quantum to each process.

Round Robin is valuable because it ensures fairness and improves responsiveness in computing systems. It is highly recommended for systems where many users or tasks need equal access to CPU resources.

Citation:

<https://www.geeksforgeeks.org/operating-systems/round-robin-scheduling-in-operating-system/>